

UNIT-III
VECTOR INTEGRATION
TUTORIAL SHEET-1

1. Find the total work done by the force represented by $\vec{F} = 3xy\hat{i} - y\hat{j} + 2zx\hat{k}$ in moving a particle round the circle $x^2 + y^2 = 4$, $x = 2 \cos \theta$, $y = 2 \sin \theta$ and $z = 0$, $0 \leq \theta \leq 2\pi$.

Ans: 0

2. Evaluate $\int_C y^2 dx - 2x^2 dy$ along the parabola $y = x^2$ from (0,0) to (2,4).

3. Evaluate the line integral $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$, where C : square: $x = \pm 1$, $y = \pm 1$

Ans: 0

4. Verify Green's theorem for $\int_C (e^{-x} \sin y)dx + (e^{-x} \cos y)dy$, where C is the rectangle, whose vertices are $(0, 0)$, $(\pi, 0)$, $(\pi, \frac{\pi}{2})$ and $(0, \frac{\pi}{2})$.

Ans: $[2(e^{-\pi} - 1)]$

5. Using Green's theorem, evaluate $\oint_C (x^2 - \cosh y)dx + (y + \sin x)dy$ where C is the boundary of the rectangle $0 \leq x \leq \pi$, $0 \leq y \leq 1$.

Ans: $\pi(\cosh 1 - 1)$

6. Using the Green's theorem, evaluate $\oint_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed by the lines $x = \pm 1$ and $y = \pm 1$.

Ans: $\frac{128}{5}$

7. If S is the surface of the sphere $x^2 + y^2 + z^2 = d^2$ and $\vec{A} = ax\hat{i} + by\hat{j} + cz\hat{k}$, evaluate $\iint_S \vec{A} \cdot \hat{n} ds$.

Ans: $\frac{4\pi d^3}{3}(a + b + c)$

8. If $\vec{F} = 2y\hat{i} - 3\hat{j} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, show that $\iint_S \vec{F} \cdot \hat{n} ds = 132$.

9. Find the surface integral over the parallelepiped $x = 0, y = 0, x = 1, y = 2, z = 3$ when $\vec{A} = 2xy\hat{i} + yz^2\hat{j} + xz\hat{k}$

Ans: 33.

10. Using divergence theorem, evaluate $\iint_S \vec{r} \cdot \hat{n} ds$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 9$.

Ans: 108π

11. Verify divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.

$$\iint_S \vec{F} \cdot \hat{n} \, ds = \frac{3}{2}$$

12. Using divergence theorem, evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ over the entire surface S of the region above xy plane bounded by the cone $x^2 + y^2 = z^2$ the plane $z = 4$ where $\vec{F} = 4xz\hat{i} - xyz^2\hat{j} + 3z\hat{k}$

Ans: 704π

13. Verify Stokes's theorem where $\vec{A} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$ and S : upper half of the surface of the sphere $x^2 + y^2 + z^2 = 1$

Ans: π

14. Evaluate $\oint_C xy \, dx + xy^2 \, dy$ by Stoke's theorem where C is the square in the xy plane with vertices $(1,0) (-1,0) (0,1) (0,-1)$.

$$\text{Ans} = \frac{4}{3}$$

15. Evaluate $\oint_C 4z \, dx - 2x \, dy + 2x \, dz$ by Stoke's theorem where C is the ellipse $x^2 + y^2 = 1, z = y + 1$.

Ans: -4π